

Aparnaa Senthilnathan

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OBJECTIVE

Early-career AI/software engineer who builds and evaluates machine-learning systems across computer vision, LLM/RAG, audio ML, and backend software. Experience spans PyTorch experimentation, retrieval evaluation, GPU workflows, and turning ML prototypes into usable applications.

TECHNICAL SKILLS

AI/ML & Systems: PyTorch, Model Evaluation, RAG, LLM Agents, Slurm, TensorBoard, FiftyOne

Languages: Python, Java, JavaScript/TypeScript, SQL

Development: Docker, Linux, CI/CD, FastAPI, REST APIs

Tools: Git, Ollama, Qdrant, ChromaDB, AWS, Amazon Bedrock

EXPERIENCE

Computer Vision Intern

May 2024 – August 2024

Kitware, Inc.

Carrboro, NC

- Evaluated YOLOX and RT-DETR on 20,000 TinyBirds training and validation images to study small-object detection, running experiments with PyTorch, Slurm/multi-GPU workflows, and TensorBoard
- Implemented the RT-DETR pipeline and compared predictions with ground-truth annotations, debugging training behavior and identifying limitations in the setup that informed follow-up testing

Artificial Intelligence Research Co-Op

July 2023 – December 2023

Griffiss Institute

Rome, NY

- Investigated whether CLIP could distinguish visually different objects sharing the same name across 20 homonym-image examples
- Compared RN50 and ViT-L/14 using cosine similarity and confusion matrices, finding stronger image-image than image-text matching and similar behavior across both architectures

PROJECTS

MAARA: Multi-Agent Automated Repair & Refinement Assistant | *Team Project*

April 2026 – May 2026

- Collaborated on a team of 3 to develop a five-agent LLM software-repair pipeline to address failures of single-pass code generation; contributed to architecture and evaluation across planning, pre-generation validation, RAG-grounded coding, critique, and repair with locally served Ollama models
- Extended MAARA with a 20-query retrieval benchmark across 29 source-code chunks, comparing dense, BM25, and hybrid RRF retrieval; dense ranked the relevant chunk first for 19/20 queries and achieved 0.963 MRR vs. 0.877 hybrid and 0.727 BM25

Mood-y: Voice-Based Mood Music Recommender | *Team Project*

March 2026 – May 2026

- Co-developed an end-to-end system that turns voice input into mood-based music recommendations on a team of 3, extracting 112-dimensional acoustic features with Librosa and connecting emotion classification to a FastAPI/React application
- Compared SVM and Random Forest classifiers across eight RAVDESS emotions; the best SVM reached 91.7% accuracy and 0.916 F1, with recommendations ranked using emotion profiles, cosine similarity, and weighted audio features

FrequencyPrint: An Automated Deepfake Detection System | *Capstone Project*

January 2026 – May 2026

- Investigated whether CNN-based audio models could distinguish authentic music from AI-generated/deepfake music, comparing SimpleCNN, ResNet18, and ResNet34 with PyTorch, Librosa, and spectrogram-based inputs
- Found that initial 99–100% accuracy failed to generalize to AI-generated covers, traced the problem to dataset mismatch/overfitting, and redesigned the evaluation with more representative real and AI-cover data, reaching a more realistic best accuracy of 61.5%

EDUCATION AND CERTIFICATIONS

Rochester Institute of Technology

Rochester, NY

Bachelor of Science & Master of Science in Computer Science, RIT Presidential Scholar

August 2021 – May 2026

Certifications:

IBM SkillsBuild Data Analytics Certificate – IBM, 2026

Google AI Essentials – Google, 2025